

Khanh Nguyen, Postdoc, Ph.D.

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EDUCATION

Doctor of Philosophy in Smart Vehicle Engineering

Konkuk University, Seoul, Republic of Korea | 2022 – 2026 | **GPA: 4.0/4.0**

Dissertation: *Scaling-based design approach for tailbeat fish-inspired robots swimming at high speed and propulsive efficiency*

Master of Science in Smart Vehicle Engineering

Konkuk University, Seoul, Republic of Korea | 2019 – 2021 | **GPA: 3.9/4.0**

Thesis: *Investigation of stability and aerodynamic performance of a flapping-wing micro air vehicle in hover using 3D computational fluid dynamics analyses*

Bachelor of Engineering in Mechanical and Aerospace Engineering

Vietnam National University, Ho Chi Minh City University of Technology, Vietnam | 2013 – 2018 | **GPA: 3.2/4.0**

The France–Vietnam Training Program of Excellent Engineers (PFIEV) is accredited by France's Engineering Degree Commission and designated as a Master Program by the European Network for Accreditation of Engineering Education.

Thesis: *Computational approach on the aerodynamics of UAV combining fixed wing and three propellers* (Grade: 9.1/10)

RESEARCH EXPERIENCE

Postdoctoral Researcher - Konkuk University (03/2026 - Present)

Analyzed fast forward-flight wing kinematics with a tilted stroke plane under various inflow velocities by a wind generator.

Validated the proposed scaling-based design approach by developing a larger fish robot capable of swimming at over 2.2 m/s.

Senior Research Co-Supervisor - Ho Chi Minh City University of Technology (01/2026 – Present)

Optimized flexible flapping-foil kinematics to enhance force control authority.

Studied LEV and TEV interactions and their potential role in stall-flutter suppression in corrugated flapping wings.

Investigated the effect of high-frequency trailing-edge vibration on drag reduction and flapping-foil efficiency.

Conceptualized a bio-inspired robotic glider based on the flight characteristics of Javan cucumber seeds.

Research Assistant - Konkuk University (03/2019 – 02/2026)

Developed a fish robot capable of efficiently swimming at 1.79 m/s using a scaling-based design approach.

Investigated characteristics of dynamic flight stability and aerodynamics of a flapping-wing hovering on Mars.

Analyzed the feasibility of the gliding flight in a flying-fish-like robot after water exit.

Estimated body drag of undulatory robotic fish using the measured swimming kinematics.

Investigated the aerodynamic effects of spanwise corrugation, wing rotation angles, and clap-fling mechanism.

Compared the longitudinal and lateral stability characteristics of two flapping-wing robots.

Analyzed laminar purple separation along chordwise and spanwise positions.

Research Assistant – Ho Chi Minh City University of Technology (02/2018 - 02/2019)

Estimated the tip-path-plane angle of the forward-flight tricopter using CFD-derived aerodynamic force coefficients.

Analyzed aerodynamics of a tricopter frame during takeoff using the Virtual Blade Method in OpenFOAM.

SERVICES & MENTORSHIP

Professional Services:

Journal referee, Chinese Journal of Aeronautics (Q1), Journal of Fluids and Structures (Q1), Bioinspiration & Biomimetics (Q1), Engineering Research Express (Q2), Journal of Aeronautics Astronautics and Aviation (Q3), International Journal of Intelligent Unmanned Systems (Q4).

Academic Services & Teaching Assistant

Grading Assistant, Basics of Mechanics & Finite Element Method (Konkuk University); Fluid Mechanics (HCMUT).
Thesis Evaluator, Bachelor's graduation theses (HCMUT).

Research & Mentorship Assistant

Co-supervisor, research and capstone projects, Ho Chi Minh City University of Technology (2025-present, 2018).
Mentor, CFD simulation and mechanical design for robotic fish, Konkuk University (2026, 2024).

AWARDS & HONORS

Outstanding Dissertation Award, Korean Society of Mechanical Engineers (Top 5 nationwide; only international awardee; 2026)
First Prize, Academic Award for Outstanding Ph.D. Graduate, Engineering Division, Konkuk University (2026)
Best Paper Award, International Conference on Intelligent Unmanned Systems (2022, 2025)
Best Paper Award, Korea Society for Aeronautical and Space Sciences Conference (2024)
Excellent Student, Ho Chi Minh City University of Technology (150% Tuition Award; 2018)
Quintessential Student, Ho Chi Minh City University of Technology (125% Tuition Award; 2017, 2014)
Quintessential Student, Tran Phu High School (National University Entrance Exam; Score: 25/30, 1% Nationwide; 2013)

FELLOWSHIPS & ACADEMIC APPOINTMENTS

Brain Korea 21 Postdoctoral Scholarship, Korean Government (\$23,000; 2026)
Postdoctoral Fellowship, Konkuk University (\$33,000; 2026)
Ph.D. Fellowship, Konkuk University (\$78,000; 2022 – 2026)
Research Assistant, Ho Chi Minh City University of Technology (2018); Konkuk University (2019 – 2021). Total: \$23,200
Teaching Assistant, Ho Chi Minh City University of Technology (2018); Konkuk University (2024). Total: \$10,100

TECHNICAL SKILLS

Languages: C, Python, HTML/CSS.
Development Tools: MATLAB, Visual Studio, VS Code.
Simulation and Post-Processing Tools: ANSYS-Fluent, CFD-Post, OpenFOAM, ParaFoam.
Tools: CNC & 3D Printing Machines, Direct Linear Transformation Digitizing Tool.
Software: AutoCAD, Adobe Photoshop, SolidWorks.
Processes: Silicone Mold Making, Transducer Measurement, Image Processing.

PUBLICATIONS

Journal Papers

1. **Khanh Nguyen**, Hoon Cheol Park. Scaling-based design approach for tailbeat fish-inspired robots swimming at high speed and propulsive efficiency. Submitted to Ocean Engineering, September 2025; under 4th-round review.
2. Giheon Ha, **Khanh Nguyen**, Jiyeon Yu, Teasam Kang, Hoon Cheol Park. Aerodynamic characteristics of flapping wings at takeoff under very low air density and low gravity. Acta Astronautica, 2026. (Q1, IF = 3.7, Rank: 10/59 in Aeros. Engineering)
3. Thi Hong Hieu Le, **Khanh Nguyen**, Vuong Thi Hong Nhi. Numerical analysis for aerodynamic characteristics of the unmanned aerial vehicle (UAV) in forward flight. J. Aeronaut. Astronaut. Aviat. 2024. (Q3, IF = 1.1)
4. **Khanh Nguyen**, Giheon Ha, Teasam Kang, Hoon Cheol Park. Analysis of hovering flight stability of an insect-like flapping-wing robot in Martian condition. Aerosp. Sci. Technol. 2024. (Q1, IF = 5.8, Rank: 6/55 in Aerospace Engineering)
5. **Khanh Nguyen**, Hoon Cheol Park. Feasibility study on mimicking the tail-beating supported gliding flight of flying fish. Ocean Engineering 2023. (Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering)
6. Tan Hanh Pham, **Khanh Nguyen**, Hoon Cheol Park. A robotic fish capable of fast underwater swimming and water leaping with high Froude number. Ocean Engineering 2023. (Q1, IF = 4.6, Rank: 2/25, Top 8% in Marine Engineering)

7. **Khanh Nguyen**, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park. Comparative dynamic flight stability of insect-inspired flapping-wing micro air vehicles in hover: Longitudinal and lateral motions. *Aerosp. Sci. Technol.* 2021. (Q1, IF = 5.5, Rank: 2/34, Top 6% in Aerospace Engineering)
8. **Khanh Nguyen**, Loan Thi Kim Au, Hoang-Vu Phan, Soo Hyung Park, Hoon Cheol Park. Effects of wing kinematics, corrugation, and clap-and-fling on aerodynamic efficiency of a hovering insect-inspired flapping-wing micro air vehicle. *Aerosp. Sci. Technol.* 2021. (Q1, IF = 5.5, Rank: 2/34, Top 6% in Aerospace Engineering)
9. Doan Kim Khanh Tran, **Khanh Nguyen**, Thi Hong Hieu Le, Ngoc Hien Nguyen. Numerical simulation for the forward flight of the tri-copter using virtual blade model. *J. Adv. Res. Fluid Mech. Therm. Sci.* 2020. (Q3, IF = 0.3)

Conference Papers

1. Giheon Ha, **Khanh Nguyen**, Hoon Cheol Park, A study on the takeoff of an insect-like flapping-wing system under low air density and low gravity conditions, *Proceedings of KSAS, Korea*, 2024. (Best paper award)
2. **Khanh Nguyen**, Giheon Ha, Hoon Cheol Park, Design and fabrication of high-thrust tail-beating mechanism for fish-inspired swimming robot, *ICIUS, Indonesia*, 2024. (Presenter)
3. **Khanh Nguyen**, Hoon Cheol Park, Analytical and experimental performance verifications of a fast-swimming robotic fish, *ICIUS, Indonesia*, 2024. (Presenter)
4. **Khanh Nguyen**, Kang Teasam, Hoon Cheol Park, Hovering characteristics of an insect-like flapping-wing robot on Mars, *Proceedings of KSAS, Korea*, 2023. (Presenter)
5. **Khanh Nguyen**, Giheon Ha, Hoon Cheol Park, Preliminary design of a fish-like fast robot by scaling of the KUFish, *ICIUS, AU*, 2023.
6. **Khanh Nguyen**, Hoon Cheol Park, Roles of hydrodynamic forces generated by tail-beating motion in gliding flight of flying-fish-mimicking robot, *ICIUS, Adelaide, Australia*, 2023. (Presenter)
7. **Khanh Nguyen**, Tan Hanh Pham, Hoon Cheol Park, Numerical investigation of hydrodynamics for a fish-like robot under undulatory forward swimming, the Korean Society of Mechanical Engineers Annual Meeting, Jeju, Korea, 2022. (Presenter)
8. Tan Hanh Pham, **Khanh Nguyen**, Hoon Cheol Park, Leaping out of water of the KUFish: Prediction and demonstration, *ICIUS, Japan*, 2022. (Best paper award).
9. **Khanh Nguyen**, Tan Hanh Pham, Hoon Cheol Park, Numerical estimation of hydrodynamic thrust using the measured tail-beating kinematics of a fish-like robot, *ICIUS, Japan*, 2022. (Presenter)
10. **Khanh Nguyen**, Loan Thi Kim Au, Hoang Vu Phan, Hoon Cheol Park, Wing kinematics modulation in an insect-like tailless flapping wing micro air vehicle for higher aerodynamic efficiency, *ICIUS, Vietnam*, 2021. (Presenter)
11. **Khanh Nguyen**, Loan Thi Kim Au, Hoon Cheol Park, Three-dimensional wing kinematics for improved aerodynamic performance of insect-like flapping-wing micro air vehicle, *KSAS, Korea*, 2020. (Presenter)
12. Loan Thi Kim Au, **Khanh Nguyen**, Soo Huyn Park, Hoon Cheol Park, Effect of wing corrugation on aerodynamic performance in 3D flapping wings, *Proceedings of KSAS, Korea*, 2019. (Presenter)
13. Doan Kim Khanh Tran, **Khanh Nguyen**, Thi Hong Hieu Le, Numerical simulation for the forward flight of the tri-copter using Virtual Blade Model, *SAWAE, Malaysia*, 2019.
14. **Khanh Nguyen**, Ngoc Hien Nguyen, Thi Hong Hieu Le, Numerical approach for the vertical take-off and landing UAVs using the virtual blade model, *SAWAE, Thailand*, 2018. (Presenter)